

COM S 127 - Lab Week 3 Grading Rubric

This lab was assigned for week 3 of the Fall semester.

This lab is due by 11:59 p.m. of the sixth day after the student's initial lab meeting day.

Lab Objective

The purpose of this lab is for students to continue familiarizing themselves with important concepts such as variables, operators, the built in math functions, and the math library.

Instructions

Folder for Submission:

All the screenshots and scripts should be bundled together in a zip file with the following name:

labWeek3<netid>

For example, if your <netid> is student123, then your folder should be called:

labWeek3student123

Deliverables:

Please provide a screenshot/ screenshots of the table of contents of the Runestone textbook, with green 'check marks' next to the relevant sections noted in the **Readings** section of this document.

Create a script called pythonStatementResults.py which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called exercise1.py which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called exercise2.py which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called exercise3.py which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called exercise4.py which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called exercise5.py which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Important Notes:

Screenshots, when possible, should include some kind of identifying information unique to the student - in this case their ISU netid. For example, in Runestone, it is possible to click the 'Account' button to cause a 'flyout' with a username to be displayed. If this username is not the student's netid, the student may also take a screenshot and then write/ type their netid into the image using an image editing program such as MS Paint, paint.net, etc. This should be done in a large conspicuous font.

All scripts should include the student's name and the date they programmed the script on the top line of the file. The student should also include the week of the lab, and the exercise number on the second line. Example:

```
# <Student Name>           <The Date>
# Lab Week <week of the lab> - Exercise #<exercise number>

# Matthew Holman           9-1-2022
# Lab Week 3 - Exercise #2
```

Readings:

variables

- 2.4. Variables
- 2.10. Reassignment
- 2.11. Updating Variables

keywords

- 2.5. Variable Names and Keywords

types (basic types)

- 2.1. Variables, Expressions and Statements
- 2.2. Values and Data Types

types (type conversion functions)

- 2.3. Type conversion functions

input function

- 2.8. Input

statements and expressions

- 2.6. Statements and Expressions

operators (+, -, *, /, //, **, %)

- 2.7. Operators and Operands
- 2.9. Order of Operations

Lab Challenge Activities:

Create a single script called `pythonStatementResults.py`, which **prints** out the results of the following Python statements (as per slides 2-6 in the `labWeek3MathReview.pptx` file). To facilitate learning, students should attempt to determine the output by hand before programming/ running the script. Students should also should print out their initial hand computed answer, as well as the answer the computer provides. See the **Example Output** section for details. Please note - **all** of these exercises should appear in the `pythonStatementResults.py` file.:

- `min(1,2,3,4,5) == 7%2`
- `float(round(10.4,0))`
- `pow(2,-2) == pow(2,.25)`
- `pow(float(round(3.4,0)),float(round(3.4,0)))`
- `pow(max(1,2,3),min(4,5,6))`

Create a script called `exercise1.py`, such that it completes the task in **Exercise 1** on slide 7 of the `labWeek3MathReview.pptx` file.

Create a script called `exercise2.py`, such that it completes the task in **Exercise 2** on slide 8 of the `labWeek3MathReview.pptx` file.

Create a script called `exercise3.py`, such that it completes the task in **Exercise 3** on slide 9 of the `labWeek3MathReview.pptx` file.

- **NOTE:** Students can assume that the input the TAs enter will be congruent with what is asked for in the question. Meaning - the use of 'if' statements/ conditionals/ input validation is not required.

Create a script called `exercise4.py`, such that it completes the task in **Exercise 4** on slide 10 of the `labWeek3MathReview.pptx` file.

Create a script called `exercise5.py`, such that it completes the task in **Exercise 5** on slide 11 of the `labWeek3MathReview.pptx` file.

Resources:

Math Module Reference: <https://docs.python.org/3/library/math.html>

Example:

```
# Matthew Holman          9-1-2022
# Lab Week 3 - Math Module example

import math

print(math.sqrt(17))
```

Review the resources below to learn how to take screenshots and make zip files:

How to take screenshots (PC/ Mac): <https://www.webwise.ie/parents/take-screenshot/>

How to make zip files (PC/ Mac): <https://www.wikihow.com/Make-a-Zip-File>

Files Provided

- labWeek3MathReview.pptx

Example Code/ Output

pythonStatementResults.py code:

```
# Statement: min(1,2,3,4,5) == 7%2
print("Statement: min(1,2,3,4,5) == 7%2")

# Initial hand computer answer:
print("Hand Computer Answer: True")

# Computer generated answer:
print("Computer's Answer:", min(1,2,3,4,5) == 7%2)
```

pythonStatementResults.py output:

```
Statement: min(1,2,3,4,5) == 7%2
Hand Computer Answer: True
Computer's Answer: True
```

Exercise 4 output:

```
Size of base: 10
Size of height: 4
Area: 20.0
```

Exercise 5 output:

```
Input Class 1: Computation
Grade: 90
Input Class 2: Physics
Grade: 60
Input Class 3: Electronics
Grade: 100
Input Class 4: Mechanics
Grade: 70
The Average grade between Computation Physics Electronics and Mechanics is 80
```

Grading Items

- (Attendance) NOTE: There is no attendance grade for this week
- (Runestone Academy Reading) Has the student demonstrated that they have completed their Runestone Academy reading with a screenshot/ screenshots of the table of contents with green 'check marks' next to the relevant completed sections?: _____ / 5
- (pythonStatementResults.py) Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section? _____ / 3
- (exercise1.py) Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section? _____ / 4
- (exercise2.py) Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section? _____ / 4
- (exercise3.py) Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section? _____ / 5

- (exercise4.py) Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section? _____ / 4
- (exercise5.py) Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section? _____ / 5

TOTAL _____ / 30